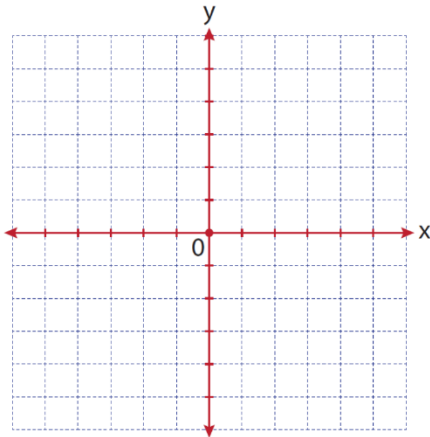


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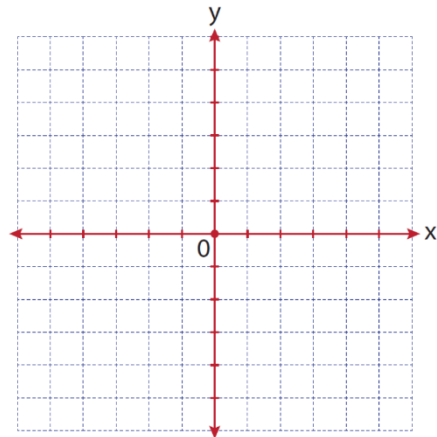
Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	TOTAL
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Q.1) (16 pts) Please provide graph sketches of the corresponding relationships

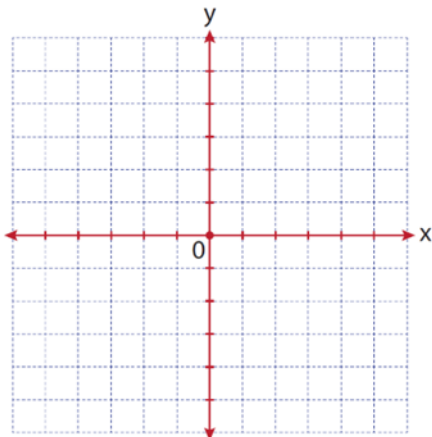
a) Linear (4 pts)



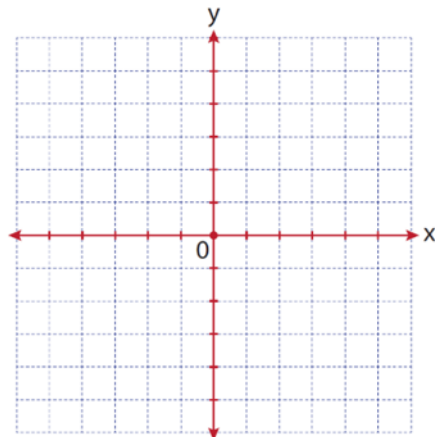
b) Absolute Value (4 pts)



c) 3rd Degree Polynomial (4 pts)



d) Normal (Gaussian) Distribution (4 pts)



Q.2) (14pts)

You are analyzing the delivery times (in minutes) for a local courier service. The following data set represents the time taken for 11 different deliveries:

Dataset: 12, 15, 17, 18, 20, 22, 23, 25, 26, 30, 52

a) (2pts) Please find the Median

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b) (2pts) Calculate the First Quartile (Q_1) and the Third Quartile (Q_3)

c) (4pts) Calculate Interquartile Range (IQR)

d) (6pts) Calculate the Lower Fence and the Upper Fence and identify the outlier(s)

Hint: **Lower Fence** ($Q_1 - 1.5 \times IQR$) and **Upper Fence** ($Q_3 + 1.5 \times IQR$)

Q.3) (25pts) A data science startup tests whether a person's primary fitness goal is associated with their preferred cuisine type. They believe that "Muscle Gain" seekers might prefer protein-heavy Indian cuisine, while "Endurance" athletes might prefer Mediterranean diets.

You collect data from 200 gym members in Istanbul and organize the observed frequencies onto the following contingency table:

Fitness Goal \ Cuisine	Indian (Protein-Rich)	Mediterranean (Balanced)	Total
Muscle Gain	70	30	100
Endurance	40	60	100
Total	110	90	200

Please answer the questions regarding the exhibit:

a) (5pts) Please create Null (H_0) and Alternative Hypothesis (H_a)

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b) (5pts) Please identify and elaborate which Hypothesis testing method can be used.

c) (5pts) Please calculate the Expected values (E_{ij}) for Muscle Gain/Indian and Endurance/Mediterranean cells

Hint: $E_{ij} = \frac{(\text{row total} \times \text{column total})}{\text{grand total}}$

d) (5pts) Please calculate statistic for the Muscle Gain – Indian cell

Hint: $\sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$

e) (5pts) Please interpret and compare the results with the critical value of 3.84 at $\alpha = 0.05$

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Q.4) (15pts) You are consulting for a food truck business in Istanbul. You have developed a Multiple Linear Regression model to predict Daily Revenue (Y, in TL) based on two predictors:

X_1 : Average Temperature (in Celsius)

X_2 : Number of Active Promotions (Discount codes sent via SMS)

The resulting regression equation from your analysis is:

$$\hat{Y} = 1500 + 120X_1 + 450X_2$$

Please answer the following questions regarding the scenario:

a) (5pts) What does the value **1500** represent in the context of the food truck's daily revenue? Please elaborate.

b) (5pts) If the temperature increases by **1 degree Celsius** while the number of promotions remains constant, what is the expected change in Daily Revenue?

c) (5pts) The owner wants to increase revenue by exactly **1,800 TL** today. If the temperature is fixed, how many **additional promotions** (X_2) must they run to hit this specific target?

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Q.5) (21pts) You are a Senior Engineer leading a technical sustainability project. You need to create a simple linear regression model how **Ambient Temperature (X, in °C)** impacts the **Charging Efficiency (Y, as a %)** of a new high-speed EV station.

You have collected the following measurements from the field:

Observation	Temperature (X)	Efficiency (Y)
1	10	95
2	15	85
3	20	80
4	25	75
5	30	65

a) (7pts) Calculate the Slope (β_1)

Hint: $\bar{X} = \frac{\sum X}{n}$ and $\beta_1 = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2}$

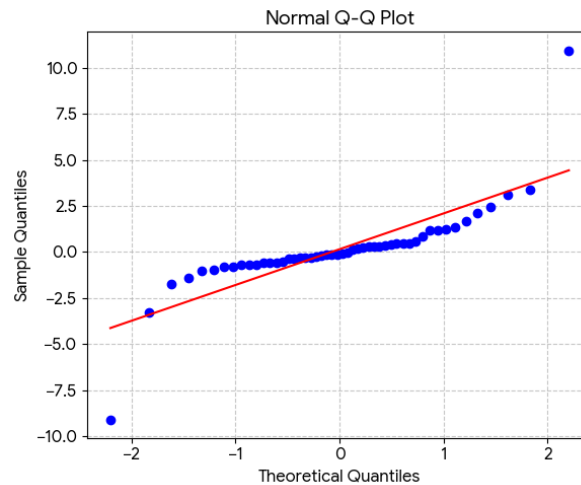
b) (7pts) Calculate the Slope (β_0)

Hint: $\beta_0 = \bar{Y} - \beta_1 \bar{X}$

c) (7pts) Please identify the final equation

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Q.6) (9pts) After building your regression model for the EV charging station efficiency, you generate a Q-Q (Quantile-Quantile) Plot of the residuals to check the assumption of Normality as follows:



Please answer the following questions referring to the exhibit:

a) (4pts) Based on the plot above, do the residuals follow a normal distribution? Why? Please elaborate.

b) (5pts) As a Data Scientist, why is it risky to rely on the **p-values** and **Confidence Intervals** provided by the regression output if the Q-Q plot looks like this?